# hermod+

## _travelers guide
## _for talking to machines

**16** tracks · 8 cv + 8 midi | **16** sequences / project | **8+4** fx · mod lanes / track

`[ step ]` press  `[ step ]` hold  turn / push `[ encoder ]`  `[ pad ]` 1 of 16
`[ tr 1 ]`–`[ tr 8 ]` track switches  › then

modes: `[ step ]` · `[ effects ]` · `[ track ]` · `[ seq ]` · on air

effects › **`[ note fx ]`** : arpeggiator · chance · echo · euclid · glide · harmonizer · hold* · randomizer · ratchet · register* · scale · swing  **`[ cv fx ]`** : envelope · lfo  **`[ converters ]`** : mod to note · note to cc  **`[ note or cv ]`** : bernoulli* · filter · midi output*  * = new in os 3.0

red = hold

### 01 anywhere — hrm_01

| | |
|---|---|
| `[ tr 1–8 ]` | select track |
| `[ play ]` | start / stop (stop returns to top) |
| `[ seq ]` + `[ pad ]` | launch that sequence |
| `[ seq ]` + `[ play ]` | restart every pattern player |
| `[ step ]` + `[ rec ]` | **on air**: pads become a keyboard |
| `[ play ]` | **bpm menu**: turn `[ encoder ]` for tempo, tap `[ rec ]` to tap tempo, sync input inside |
| `[ rec ]` | **record settings** |
| `[ step ]` `[ effects ]` … | hold any mode button to peek, release to return |

### 02 recording — hrm_02

| | |
|---|---|
| `[ rec ]` | arm overdub (default) on the selected track; starts with `[ play ]` or live while playing |
| `[ x ]` + `[ rec ]` | hard rec: overwrites old notes * |
| `[ y ]` + `[ rec ]` | looper rec: length set when you stop * |
| `[ x ]` `[ y ]` + `[ rec ]` | multitrack rec: all routed tracks * |

* factory defaults. reassign: record settings › shortcut x / y / x+y — hard rec, multitrack, looper, punch-in or nothing.

### 03 power-on — hrm_03

| | |
|---|---|
| `[ step ]` at boot | skip autoload, start a fresh project |
| `[ x ]` at boot | calibrate `[ cv 1–8 ]` outputs (voltmeter; `[ tr ]` picks voice, `[ x ]` saves) |
| `[ y ]` at boot | auto-calibrate `[ cv a–d ]` inputs from `[ cv 1 ]` (do outputs first) |

### __fig.01 : pad matrix

indexed 01–16

| | | | |
|---|---|---|---|
| 01 | 02 | 03 | 04 |
| 05 | 06 | 07 | 08 |
| 09 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

■ red = modmatrix attenuverter pads (`[ effects ]` › `[ y ]`) · `[ pad ]` n = sequence n (seq) or step n (step)

### 04 step · length / view — hrm_04

| | |
|---|---|
| `[ x ]` / `[ y ]` | page right / left (pauses follow-playhead until you leave step) |
| `[ x ]` + turn `[ encoder ]` | zoom: /4, /2, 2/3, ×1, 3/2, ×2, ×4, ×8 |
| `[ y ]` + turn `[ encoder ]` | pattern length, 16–256 steps |
| `[ y ]` + push+turn `[ encoder ]` | length in single 1/16 steps (polymeter) |
| `[ x ]` `[ y ]` + turn `[ encoder ]` | double / halve length |
| hold + turn `[ encoder ]` | rotate all events left / right |

### 05 step · notes — hrm_05

| | |
|---|---|
| `[ pad ]` | add / delete a note |
| turn `[ encoder ]` | set pitch for the next note |
| `[ pad ]` + turn `[ encoder ]` | edit note; push `[ encoder ]` cycles pitch › length › velocity |
| `[ pad ]` + push+turn `[ encoder ]` | fast scroll (pitch jumps octaves) |
| `[ pad ]` `[ pad ]` | hold several to edit a range together |
| `[ pad ]` + `[ pad ]` | copy a step, paste onto another |
| `[ pad ]` + `[ tr 1–8 ]` | quick note length |
| `[ step ]` + turn → `[ encoder ]` | **row edit** on (one pitch per row); turn ← exits |
| `[ step ]` in row edit | cycle through pitches used in the pattern |
| `[ midi ]` / on air chord | play it, then `[ pad ]` drops the whole chord in (note learn) |

### 06 step · mod / gate — hrm_06

| | |
|---|---|
| `[ step ]` | toggle mod / gate tab |
| `[ step ]` + turn `[ encoder ]` | switch mod value ← → interpolation (off, lin, s, log, exp) |
| `[ pad ]` + turn `[ encoder ]` | offset that mod value / gate length |
| `[ tr ]` again | multi-voice layout: opens that voice's gate view |

### 07 step · analyzer — hrm_07

hold `[ step ]` and press a track switch to show and edit that event lane:

| `[ tr 1 ]` | `[ tr 2 ]` | `[ tr 3 ]` | `[ tr 4 ]` |
|---|---|---|---|
| notes | pitchbend | aftertouch | sustain |
| `[ tr 5 ]` | `[ tr 6 ]` | `[ tr 7 ]` | `[ tr 8 ]` |
| mod | mod 2 | mod 3 | mod 4 |

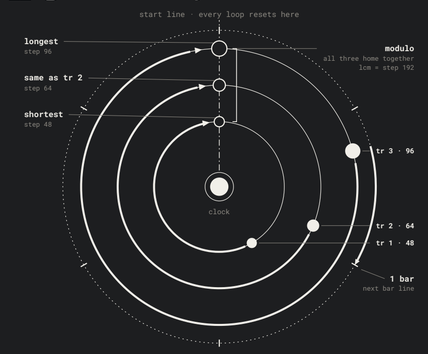
| | |
|---|---|
| `[ step ]` + `[ x ]` | erase the shown lane in this pattern (row only, in row edit) |
| `[ step ]` + `[ y ]` | randomize the shown lane |

### 08 track — hrm_08

| | |
|---|---|
| `[ track ]` or turn `[ encoder ]` | flip cv tracks 1–8 ← → midi tracks 1–8 |
| `[ pad ]` | mute track in this pattern |
| `[ track ]` + `[ pad ]` | mute track in all patterns |
| `[ x ]` | layout configurator (voices › layout) |
| `[ y ]` | global settings |
| push `[ encoder ]` | copy, paste, clear, rename track |
| `[ tr 1–8 ]` | per-track config (see hrm_12) |

### __fig.02 : the orrery

mm_01




the first planet home launches it · modulo waits for all three

### 09 effects — hrm_09

| | |
|---|---|
| empty `[ pad ]` | jump into "add effect" |
| filled `[ pad ]` | jump into "edit effect" (any pad exits) |
| `[ pad ]` | mute effect in this pattern |
| `[ effects ]` + `[ pad ]` | mute effect in all patterns |
| `[ effects ]` again | toggle parameter view ← → fx rack |
| `[ encoder ]` on slot | context menu: edit, copy, paste, delete (paste works track to track) |
| `[ encoder ]` on param | global value ← → **pattern-only value** |
| `[ y ]` | **modmatrix**, 4 lanes; `[ pad ]` picks a lane |
| `[ pad 2 · 6 · 10 · 14 ]` | lane's attenuverter / polarity / offset (fig. 01) |

### 10 seq — hrm_10

| | |
|---|---|
| `[ pad ]` | launch sequence (waits for sync) |
| `[ x ]` + `[ pad ]` | append to song chain |
| `[ x ]` tap | clear the chain |
| `[ x ]` + turn `[ encoder ]` | sync: modulo, bars, shortest / longest, same as tr; push+turn `[ encoder ]` = step precise |
| `[ y ]` + turn `[ encoder ]` | run: sync / instant, restart / free |
| live `[ pad ]` + empty `[ pad ]` | duplicate the playing sequence |
| turn › push `[ encoder ]` | chain stopped: jump to any chain step |
| `[ seq ]` `[ seq ]` | straight to project save / load |

### 11 on air — `[ step ]` + `[ rec ]` — hrm_11

| | |
|---|---|
| turn `[ encoder ]` | lowest note on the pads |
| `[ x ]` / `[ y ]` | octave shift |
| `[ rec ]` | record what you play |
| mod track | pads become 16 fixed mod values |
| exit | `[ step ]` or any mode button |

### __fig.03 : front panel

01–11 = sections · 13–18 = hardware

`[ 01 ]` **__play :** hold play = bpm
`[ 02 ]` **__rec :** hold rec = rec settings
`[ 03 ]` **__x :** zoom, rack, voices, song
`[ 04 ]` **__encoder :** turn / push
`[ 05 ]` **__pad :** matrix, 16 pads
`[ 06 ]` **__cv 1–8 __gate 1–8**
`[ 07 ]` **__tr 1–8 :** track switches
`[ 08 ]` **__track**
`[ 09 ]` **__effects**
`[ 10 ]` **__seq**
`[ 11 ]` **__step**

`[ 13 ]` **__cv a–d :** ins · a/b, c/d
`[ 14 ]` **__usb device :** usb in
`[ 15 ]` **__usb host :** out
`[ 16 ]` **__midi :** in · out
`[ 17 ]` **__reset __clock :** outs
`[ 18 ]` **__y :** length, matrix, settings, run




## _where things hide

12 · hrm_12

### global settings `[ track ]` › `[ y ]`

- **sync in** clock source, `[ cv a–d ]` reset / play / rec, clock div
- **sync out** clock + transport per port, `[ reset ]` mode, `[ clock ]` rate
- **midi in** active-track port / channel, filters, pc › seq
- **midi thru** port-to-port copies
- **cv/gate in** `[ cv a–d ]` routing, range, pitch offset
- **misc** autoload, leds, hold time, smart overdub, follow playhead
- **rec** same as hold `[ rec ]`
- **tuner** turn `[ encoder ]` to pick `[ cv a–d ]`
- **monitor** cv in › turn right: midi in › jitter
- **custom scales** 16 user scales for the scale effect
- **save** nothing above sticks until you save
- **info** os version, cpu / ram

### quick routes

- **rec options** hold `[ rec ]`
- **tempo + sync** hold `[ play ]`
- **modmatrix** `[ effects ]` › `[ y ]`
- **voice layouts** `[ track ]` › `[ x ]`
- **hold threshold** `[ track ]` › `[ y ]` › misc
- **seq copy** `[ seq ]` › push `[ encoder ]` (across projects)

### per-track config hold

`[ tr 1–8 ]`

- **player** quantize, apply sustain, transpose, mute affects input
- **inputs** midi port + channel, mod cc, `[ cv a–d ]` note / mod in, generator trigs
- **outputs** midi port + channel, mod 1–4 / pitchbend / aftertouch destinations
- **cv** `[ gate ]` retrig, poly allocator, v/oct · 1.2 v · hz/v, finetune, bend range, `[ cv ]` min / max

transpose leader = tr8 midi, always. enable player › transpose on it, then on followers.

### pattern edit `[ step ]` › push

`[ encoder ]`

- **copy / paste** whole track, or per lane (with page variants)
- **delete** per event lane
- **generator** randomize all, or just pitch / length / velocity
- **gen settings** grid, density, pitch / length / velocity ranges

### projects `[ seq ]` › push `[ encoder ]`

- **save load** save, save as, new
- **load** delete lives in here. shortcut: `[ seq ]` `[ seq ]`
- **after load** `[ y ]` take its bpm · `[ x ]` keep current. swaps in sync, no stop.

# hermod+ _travelers guide _for talking to machines

**16** tracks · 8 cv + 8 midi | **16** sequences / project | **8+4** fx · mod lanes / track

**[ step ]** press **[ step ]** hold turn / push **[ encoder ]** **[ pad ]** 1 of 16
**[ tr 1 ]**–**[ tr 8 ]** track switches › then

modes: [ step ] · [ effects ] · [ track ] · [ seq ] · on air

effects › **[ note fx ]** : arpeggiator · chance · echo · euclid · glide · harmonizer · hold* · randomizer · ratchet · register* · scale · swing **[ cv fx ]** : envelope · lfo **[ converters ]** : mod to note · note to cc **[ note or cv ]** : bernoulli* · filter · midi output* * = new in os 3.0

red = hold

## 13 cv in · setup

hrm_13

| | |
|---|---|
| **[ track ]** › **[ y ]** cv-gate in | **active track note**: cv/gate a+b or c+d → notes on the selected track, quantized c0–c10 |
| same menu | **active track mod**: cv a–d → unquantized 16-bit mod. one input can't be note and mod at once |
| **[ tr 1–8 ]** › inputs | per-track routing: **input note** a+b / c+d, **input mod** a–d. stays routed when the track isn't selected |
| cv a–d range | –5 › +5 (default) · 0 › +5 · –2.5 › +2.5 · –1 › +1. match your sequencer: **0 › +5** for 0 V-up pitch |
| gate capture | 0–20 ms wait between gate and cv read. raise it if notes land wrong (slewed or gliding cv) |
| cv pitch offset | ±60 st per input: puts the source on the right octave* |

## 14 record a sequencer

hrm_14

| | |
|---|---|
| **[ y ]** + turn **[ encoder ]** | in step: pattern length to match the sequencer's loop (see 04) |
| **[ rec ]** | overdub: keeps adding on every pass |
| **[ x ]** + **[ rec ]** | **hard rec**: replaces the selected track only. the way to swap a line |
| **[ y ]** + **[ rec ]** | **looper**: length is set when you stop — no step counting |
| **[ rec ]** punch in | waits for the first incoming note (note tracks only) |
| **[ tr ]** › player | **mute affects input** no = mute silences the recorded notes, the live sequencer still plays through |

note track = quantized notes you can edit in step. 1 voice ► modulation + input mod = raw cv, keeps glides.

## __fig.04 : the sequencer loop

patch




## 15 seq · copy + vary

hrm_15

| | |
|---|---|
| **[ pad ]** + empty **[ pad ]** | in seq: **duplicate the playing sequence**, every track (synced) |
| **[ encoder ]** in seq | copy / paste / clear a whole sequence, also across projects* |
| **[ tr 1 ]** › **[ x ]** + **[ rec ]** | on the copy: re-record the sequencer track only |
| **[ step ]** + **[ x ]** | or wipe this pattern's shown events (e.g. notes), then overdub |
| **[ encoder ]** in step | copy / paste track: one lane, full length (or one event type / page) |
| **[ pad ]** in track | pattern mute: this sequence only |
| **[ encoder ]** on fx param | pattern value: an fx setting per sequence |

seq n = pattern n of all 16 tracks.

## 16 chain sequences

hrm_16

| | |
|---|---|
| **[ x ]** + **[ pad ]** … | queue in order; repeat a pad to repeat it: 1 · 1 · 2 · 1 |
| **[ x ]** tap | erase the queue |
| **[ play ]** | the chain plays in a loop, following sync + run |
| turn › push **[ encoder ]** | chain stopped: start from any step* |
| **[ x ]** + turn **[ encoder ]** | sync: modulo · bars · shortest / longest · same as tr (push = step precise) |
| **[ y ]** + turn **[ encoder ]** | run: sync / instant · restart / free |

[ play ] again resumes from the current sequence, not the top of the chain.

## __fig.05 : the chain

seq mode




pink = the lane you re-recorded · every other track rides along untouched

## 17 note fx

hrm_17

| | |
|---|---|
| ARP arpeggiator | style · rate 1/1–1/64 · octave ±5 · gate 0–200% · re-trig (note / bars) · repeat x1–16. put **hold** first to run it hands-free |
| CHANCE chance | play / skip by %. **lot** rolls once per bar · beat · step group. chance = velocity: odds per note. **sync** + sync chance guard the downbeats |
| DELAY echo | note repeats: time · 0–16 · fade velocity / gate / pitch. pitch up and down both set = alternates |
| EUCLID euclid* | steps + fills 1–32 · rate · rotate · reset. note input **in** = plays the held notes; a fixed note passes input on, so stack several for polymeters |
| GLIDE glide* | linear · exp · smooth, 0–2500 ms then 3 – 60 s |
| HARMO harmonizer | adds up to 4 notes ±24 st. origin off drops the played note |
| HOLD hold* | hold (toggle notes) or relatch (new chord each gesture) · max notes · reset |
| RANDOM randomizer | note ±12 · octave ±5 · velocity · length · chance per note |
| RATCHET ratchet | splits each note at rate · gate · mod amount / speed stretch and shrink lengths |
| REGISTR register* | turing-style: seed · rate · bits 2–16 · chaos (0 / 100% = locked) · center · span · steady. re-seeds on unmute |
| SCALE scale | color › scale · key · stick up / down / filter / odd · trsp ±36 before correcting. after random sources |
| SWING swing | groove 50% = straight · sync (1/16 classic) · accent · human |

order matters: arp › harmonizer ≠ harmonizer › arp. two of the same effect is fine.

## __fig.06 : one modmatrix lane

effects › y




pad 2 · 6 · 10 · 14 open lanes 1–4 · the dot shows where the source sits

cv ranges per input: settings › cv-gate in

## 18 cv fx

hrm_18

| | |
|---|---|
| ENV envelope | ahdsr + curves · depth · offset. mode gate on / trig / wait ahd · sidechain any or one note · dest mod 1–4 / pb / at / cc |
| LFO lfo | sine · tri · ramp · square · rand. sync 1/64–128/1 or free 0.1 Hz – ~1 kHz · range ±100% (– inverts) · offset · phase · mute w/ track* |
| dest mod | layout 2 voices ► mono + modulation: comes out the 2nd voice cv; its gate is still free |
| mute / delete | sends the default: envelope = offset, lfo = centre |
| zipper noise | on a fast vca: slew it, or a ~1.5–2 kHz low-pass (1 kΩ + 100 nF) |

## 20 modmatrix · patch tips

hrm_20

| | |
|---|---|
| **[ effects ]** › **[ y ]** | 4 lanes per track: cv a–d, cc 1–119 or recorded mod 1–4 → any fx parameter |
| **[ pad 2 · 6 · 10 · 14 ]** | attenuverter ±100% · polarity · offset. it **adds** to the parameter, you can still turn it |
| ride → replay | record a knob or cv as mod 1–4, it drives the fx per pattern |
| self-patch | lfo / envelope out a mod voice → cv c → modmatrix → another effect's parameter |
| self-patch | a spare gate out → gate a–d set as **trig generator** ( **[ tr ]** › inputs): new pattern every hit · **trig generator mod** = new mod lane, notes kept* |

## 19 convert · route

hrm_19

| | |
|---|---|
| TOMIDI mod to note | cv / cc → notes. s&h on note-on or every 1/64–1/1 · note min / max · mono track · only while playing |
| TOCC note to cc | note / velocity / on-off → a cc · default value when idle · pass notes |
| FILTER filter | passes a note or mod range; min above max blocks it instead |
| BERNOU bernoulli* | coin-flips each event to midi path a or b by % |
| OUT midi output* | taps the chain mid-way to a midi port. before a harmonizer = plain notes out, chords on cv |

## __fig.03 : front panel

01–11 = sections · 13–18 = hardware

**[ 01 ] __play :** hold play = bpm
**[ 02 ] __rec :** hold rec = rec settings
**[ 03 ] __x :** zoom, rack, voices, song
**[ 04 ] __encoder :** turn / push
**[ 05 ] __pad :** matrix, 16 pads
**[ 06 ] __cv 1–8 __gate 1–8**
**[ 07 ] __tr 1–8 :** track switches
**[ 08 ] __track**
**[ 09 ] __effects**
**[ 10 ] __seq**
**[ 11 ] __step**

**[ 13 ] __cv a–d :** ins · a/b, c/d
**[ 14 ] __usb device :** usb in
**[ 15 ] __usb host :** out
**[ 16 ] __midi :** in · out
**[ 17 ] __reset __clock :** outs
**[ 18 ] __y :** length, matrix, settings, run




## _where things hide

12 · hrm_12

### global settings [ track ] › [ y ]

- sync in clock source, **[ cv a–d ]** reset / play / rec, clock div
- sync out clock + transport per port, **[ reset ]** mode, **[ clock ]** rate
- midi in active-track port / channel, filters, pc › seq
- midi thru port-to-port copies
- cv/gate in **[ cv a–d ]** routing, range, pitch offset
- misc autoload, leds, hold time, smart overdub, follow playhead
- rec same as hold **[ rec ]**
- tuner turn **[ encoder ]** to pick **[ cv a–d ]**
- monitor cv in › turn right: midi in › jitter
- custom scales 16 user scales for the scale effect
- save nothing above sticks until you save
- info os version, cpu / ram

### quick routes

- rec options hold **[ rec ]**
- tempo + sync hold **[ play ]**
- modmatrix **[ effects ]** › **[ y ]**
- voice layouts **[ track ]** › **[ x ]**
- hold threshold **[ track ]** › **[ y ]** › misc
- seq copy **[ seq ]** › push **[ encoder ]** (across projects)

### per-track config hold

**[ tr 1–8 ]**

- player quantize, apply sustain, transpose, mute affects input
- inputs midi port + channel, mod cc, **[ cv a–d ]** note / mod in, generator trigs
- outputs midi port + channel, mod 1–4 / pitchbend / aftertouch destinations
- cv **[ gate ]** retrig, poly allocator, v/oct · 1.2 v · hz/v, finetune, bend range, **[ cv ]** min / max

transpose leader = tr8 midi, always. enable player › transpose on it, then on followers.

### pattern edit [ step ] › push

**[ encoder ]**

- copy / paste whole track, or per lane (with page variants)
- delete per event lane
- generator randomize all, or just pitch / length / velocity
- gen settings grid, density, pitch / length / velocity ranges

### projects [ seq ] › push **[ encoder ]**

- save load save, save as, new
- load delete lives in here. shortcut: **[ seq ]** **[ seq ]**
- after load **[ y ]** take its bpm · **[ x ]** keep current. swaps in sync, no stop.